

# Audio Dequantization with a Phase-Aware Prior

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**Abstract**—Audio dequantization is the process of improving the perceived quality of an audio signal that has been degraded by quantization. A recently proposed method successfully uses a phase-aware penalty function as a prior to solve the problem of audio inpainting. The phase-aware prior promotes sinusoidality by minimizing temporal phase variation in a specific time-frequency representation based on instantaneous frequency estimation of the audio signal. To solve the dequantization problem, the original approach for inpainting is reformulated and a different projection operator is proposed. The resulting algorithm is presented and compared to an existing sparsity-based formulation in terms of the signal-to-distortion ratio and the PEMO-Q perceptual metric. The proposed method is comparable to methods based on sparsity, but further testing is required.

**Index Terms**—dequantization, instantaneous frequency, discrete Gabor transform, convex optimization

## I. INTRODUCTION

While digitizing an audio signal, there are three processes at play: sampling, quantization and coding. At least in theory, sampling and coding can be performed without a loss of information when adhering to the sampling theorem (and when the frequency range of a signal is finite). In contrast, quantization always introduces nonlinear distortion. Audio signals are usually quantized at word lengths of 16 or 24 bits per sample (bps), as the human perception cannot perceive such detail (for reference, CDs are quantized at 16 bits). As the number of bits is decreased, the distortion artifacts become audible and unpleasant. A decrease of bit depth can be desirable for a reduction of file size or an undesirable consequence of an incorrectly utilized dynamic range of recording equipment.

Quantization is understood as a mapping from a set with a larger cardinality to a set with a smaller one. This function is not injective (by the pigeonhole principle) thus its inverse function does not exist and the preimage of an observation is not a singleton. A uniform mid-riser quantization function with a quantization step  $\Delta = 2^{-w+1}$  can be expressed as

$$x^q = \text{sgn}^+(x) \cdot \Delta \cdot \left( \left\lfloor \frac{|x|}{\Delta} \right\rfloor + \frac{1}{2} \right), \quad (1)$$

where  $x$  can be thought of as a single sample or as a vector ( $\mathbf{x}$ ) in  $\mathbb{R}^L$ , where  $L$  represents the length of an audio signal in samples. The function  $\text{sgn}^+(x)$  returns  $-1$  for  $x < 0$  and  $1$  otherwise, and  $\lfloor \cdot \rfloor$  is the floor function returning the less than or equal integer. Fig. 1 is an illustration of a mid-tread uniform quantized signal with a quantization step  $\Delta$ .

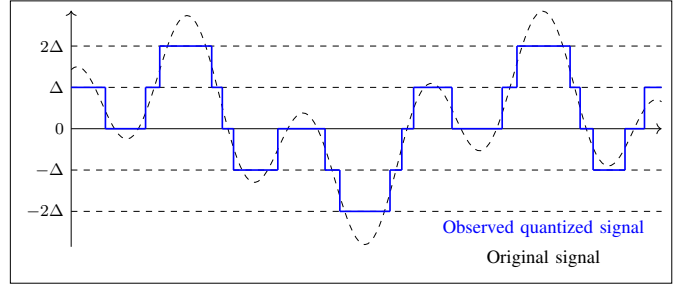


Fig. 1. Quantized signal – uniform mid-tread quantization.

The process of approximating the original signal from a quantized observation is called dequantization or bit-depth expansion. The practical goal of dequantization is finding a signal such that the perceptible difference between the original and the dequantized signal is as small as possible. The problem of finding such a signal is ill-posed as there are many possible solutions for a given signal. We denote the set of consistent solutions as

$$\Gamma = \{\mathbf{x} \in \mathbb{R}^L : \|\mathbf{x} - \mathbf{x}^q\|_\infty < \Delta/2\}. \quad (2)$$

Restricting a solution to belong to this set ensures retaining the known properties of the quantized signal (the interval where the original sample was is given by the quantized sample). Whether the dequantized signal must lie in this set divides the dequantization methods to consistent and inconsistent.

This paper is based on research [1] using instantaneous frequency to solve the problem of audio inpainting while proposing a framework of instantaneous frequency utilization for similar problems [2], [3]. Other methods used for dequantization utilize sparsity in the time-frequency (TF) domain [4], [5] and autoregressive modeling [6]. Many dequantization methods use a form of sinusoidality as a regularizer, where sinusoidality is the assumption that audio signals are composed of a relatively small number of sine waves. This results in a TF representation of an audio signal being sparse or containing components whose phase can be approximated as a linear function.

This paper is organized as follows. Section II reviews the preliminaries. Section III presents the proposed reformulation and algorithm. Section IV describes the experiments and results. Section V concludes the paper.

## II. PRELIMINARIES

### A. Time-Frequency Transformation

The time-frequency transform used in this paper is the discrete Gabor transform (DGT). It is the discrete and down-sampled version of the short-time Fourier transform similar to the one proposed by Denis Gabor [2]. This transform can be interpreted as multiplying the signal by a sliding window function and subsequent discrete Fourier transform. For a fixed window function  $w \neq 0$ , the DGT is a mapping of a signal  $\mathbf{x} \in \mathbb{C}^L$  to a matrix  $G_w \mathbf{x} \in \mathbb{C}^{M \times N}$  defined component-wise as [2]

$$(G_w \mathbf{x})[m, n] = \sum_{l=0}^{L-1} x[l] \bar{w}[l - an] e^{-2\pi i l m / M} \quad (3)$$

where we assume that the signal length  $L$  is divisible by  $M$  and  $N$ , so that  $L = Mb = Na$ . This assumption is necessary and could be handled by padding [7]. The parameter  $a \in \mathbb{N}$  is the time shift step or hop size (downsampling in the time domain),  $M$  is the number of frequency bins (channels), and  $b \in \mathbb{N}$  determines the downsampling in the frequency domain [2]. The resulting matrix of complex coefficients is called the spectrogram.

### B. Sparsity-Based Optimization

Sparsity-based dequantization aims to promote sparsity in the spectrogram of the dequantized signal. Such an approach can be formulated as the following unconstrained minimization problem [4]

$$\text{Find } \mathbf{x}^* = \arg \min_{\mathbf{x} \in \mathbb{R}^L} \lambda \|A\mathbf{x}\|_0 + \phi_\Gamma(\mathbf{x}), \quad (4)$$

where  $\mathbf{x}$  is the time domain signal,  $A$  is a time-frequency analysis operator (e.g. the DGT) and  $\Gamma$  is the set of consistent solutions. The hyperparameter  $0 < \lambda \in \mathbb{R}$  represents the trade-off between sparsity and consistency [4]. In the consistent formulations  $\phi_\Gamma$  is the indicator function  $\iota_\Gamma$  of  $\Gamma$ ; in the inconsistent formulations, it is the distance function  $d_\Gamma$  from  $\Gamma$ .

Since the formulation with the  $\|\cdot\|_0$  norm is NP-hard to solve, it is typically replaced by the  $\|\cdot\|_1$  norm, a process known as  $\ell^1$  relaxation. The formulation (4) is called the analysis formulation.

### C. Instantaneous Frequency

Instantaneous frequency (IF) is defined as the time derivative of the phase component of a signal. This value can be interpreted as the frequency of the signal at a particular instant. However, this interpretation is only approximate: the IF is evaluated over a time interval equal to the support of the window function, in accordance with the Gabor uncertainty principle.

The first-order approximation of the instantaneous frequency can be computed as

$$\omega_{\mathbf{x}} = -\text{Im} \left[ \frac{G_{w'} \mathbf{x}}{G_w \mathbf{x}} \right] = -\text{Im} \left[ \frac{G_{w'} \mathbf{x} G_w \mathbf{x}}{(G_w \mathbf{x})^2} \right], \quad (5)$$

where  $G_{w'} \mathbf{x}$  denotes the DGT of  $\mathbf{x}$  using a time differentiated window  $w'$ , and  $\text{Im}[\cdot]$  denotes the imaginary part.

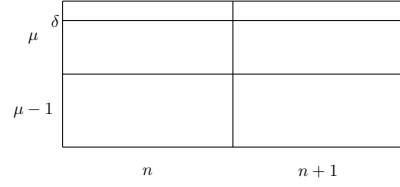


Fig. 2. Instantaneous frequency  $\delta$  in the spectrogram matrix.

### D. Phase-Aware Prior

As was established in [8], the difference between time adjacent samples of a single component complex sinusoid signal sampled with a period  $T$  is

$$x[l] \approx x[l + 1] e^{-i T \omega}, \quad (6)$$

if the amplitude is changing relatively slowly. This only translates into the TF representation if the frequency of the signal corresponds to a center of some frequency bin, which is generally not the case. Thus the bin-relative frequency shift  $\delta$  was introduced in [3]. Let the angular frequency  $\omega = 2\pi f = 2\pi(\mu + \delta)/M$  and  $l = an$ , the same signal can be represented as

$$x[an] = e^{2\pi i a n (\mu + \delta) / M}, \quad (7)$$

where  $\mu \in \{0, 1, \dots, M - 1\}$  is the center of some frequency bin and  $\delta$  is a frequency-shift from the center [1], see Fig. 2. Shifting the signal by  $a$  samples results in the following phase rotation

$$x[a(n + 1)] = e^{2\pi i a n (\mu + \delta) / M} e^{2\pi i a \mu / M} e^{2\pi i a \delta / M} \quad (8)$$

$$= x[l] e^{2\pi i a \mu / M} e^{2\pi i a \delta / M}. \quad (9)$$

Applying the DGT results in following phase shift in time-adjacent TF coefficients

$$(G_w \mathbf{x})[\mu, n + 1] = \sum_{l=0}^{L-1} x[l] e^{2\pi i a \mu / M} e^{2\pi i a \delta / M} e^{-2\pi i (l + a) \mu / M} \quad (10)$$

$$= (G_w \mathbf{x})[\mu, n] e^{2\pi i a \delta / M}, \quad (11)$$

thus the phase rotation in the TF representation is

$$(G_w \mathbf{x})[\mu, n + 1] e^{-2\pi i a \delta / M} = (G_w \mathbf{x})[\mu, n], \quad (12)$$

therefore the time adjacent samples in the spectrogram are identical up to a phase factor [1]. As the mismatch is accumulating along the time axis of the spectrogram at a known rate, the phase rotation can be eliminated through the whole spectrogram by the cumulative sum of  $\omega_{\mathbf{x}}$  as

$$(R_{\omega_{\mathbf{x}}} G_w \mathbf{x})[m, n] = (G_w \mathbf{x})[m, n] e^{-2\pi i a \sum_{t=0}^{n-1} \omega_{\mathbf{x}}[m, t] / M}. \quad (13)$$

This operation is called phase correction (compensation) [1]. The last part of the penalty function is the time difference operator  $D : \mathbb{C}^{M \times N} \rightarrow \mathbb{C}^{M \times N-1}$  defined as

$$(D\mathbf{c})[m, n] = c[m, n] - c[m, n + 1]. \quad (14)$$

This operator computes the temporal difference in the phase corrected spectrogram.

### E. Chambolle–Pock Algorithm

To solve the convex optimization problem formulated later a Chambolle–Pock algorithm [9], [10] is used. A short formal definition follows.

Let  $\mathcal{X}$  and  $\mathcal{Y}$  be real Hilbert spaces and let  $f : \mathcal{X} \rightarrow \bar{\mathbf{R}}$  and  $g : \mathcal{Y} \rightarrow \bar{\mathbf{R}}$  be convex, proper, lower semicontinuous functions. To solve a problem of the form:

$$\underset{\mathbf{x} \in \mathcal{X}}{\text{minimize}} f(\mathbf{x}) + g(L\mathbf{x}), \quad (15)$$

where  $L : \mathcal{X} \rightarrow \mathcal{Y}$  is a bounded linear operator, an Algorithm 1 is used. This algorithm weakly converges to a solution if the parameters  $\sigma > 0, \tau > 0$  satisfy  $\sigma\tau\|L\|_{op}^2 \leq 1$ , and  $(\alpha^{(i)})_{i \in \mathbb{N}}$  is a sequence in  $[0, 2]$  such that  $\sum_{i \in \mathbb{N}} \alpha^{(i)}(2 - \alpha^{(i)}) = +\infty$  [9].

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#### Algorithm 1 Chambolle–Pock algorithm

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for  $i = 0, 1, \dots$ 
 $\mathbf{x}^{[i+\frac{1}{2}]} = \text{prox}_{\tau f}(\mathbf{x}^{[i]} - \tau L^* \mathbf{u}^{[i]})$ 
 $\mathbf{u}^{[i+\frac{1}{2}]} = \text{prox}_{\sigma g^*}(\mathbf{u}^{[i]} + \sigma L(2\mathbf{x}^{[i+\frac{1}{2}]} - \mathbf{x}^{[i]}))$ 
 $\mathbf{x}^{[i+1]} = \mathbf{x}^{[i]} + \alpha^{[i]}(\mathbf{x}^{[i+\frac{1}{2}]} - \mathbf{x}^{[i]})$ 
 $\mathbf{u}^{[i+1]} = \mathbf{u}^{[i]} + \alpha^{[i]}(\mathbf{u}^{[i+\frac{1}{2}]} - \mathbf{u}^{[i]})$ 

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### III. FORMULATION

Whether it is inpainting, declipping or dequantization the problem formulation is of the same form. The optimization problem is to find

$$\mathbf{x}^* = \arg \min_{\mathbf{x} \in \mathbb{R}^L} \lambda \|DR_{\omega_y} G_w \mathbf{x}\|_1 + \iota_\Gamma(\mathbf{x}), \quad (16)$$

where  $\Gamma$  is the corresponding set of consistent solutions and  $\omega_y$  is the (relative) instantaneous frequency of the observed signal [1]. The indicator function of a convex set and function representing the  $\ell^1$  norm are convex, proper l.s.c functions. Thus the Chambolle–Pock algorithm (Alg. 1) can be applied to this minimization problem resulting in Algorithm 2 called U-PHADQ. Note that the update of the IF occurs only outside of the CP (inner) loop.

When solving the problem of dequantization the projection function is defined component-wise as

$$(\text{proj}_\Gamma \mathbf{x})[n] = \min \left( \max \left( x[n], x^q[n] - \frac{\Delta}{2} \right), x^q[n] + \frac{\Delta}{2} \right)$$

where  $x^q$  is the observed quantized sample and  $x$  is the sample to be projected. As a vector operator this could be imagined as an intersection of hyperrectangles corresponding to the quantization intervals of each sample. Since each set is convex, a finite intersection is also a convex set.

The soft thresholding operator is the proximal operator of the  $\|\cdot\|_1$  norm defined as [1], [5]

$$\text{soft}_\lambda(x) = \text{sgn}(x) \max(|x| - \lambda, 0). \quad (17)$$

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#### Algorithm 2 U-PHADQ

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Initialize  $\mathbf{x}^{[0]} \leftarrow 0, \hat{\mathbf{x}}^{[0]} \leftarrow 0, \mathbf{u}^{[0]} \leftarrow 0, J > 0, I > 0$ 
for  $j = 0, \dots, J - 1$  do
 $\omega^{[j]} = -\text{Im}(G_w^* \hat{\mathbf{x}}^{[j]} \oslash G_w \hat{\mathbf{x}}^{[j]})$ 
for  $i = 0, \dots, I - 1$  do
 $\mathbf{x}^{[i+\frac{1}{2}]} = \text{proj}_\Gamma(\mathbf{x}^{[i]} - \tau G_w^* R_{\omega^{[j]}}^* D^* \mathbf{u}^{[i]})$ 
 $\mathbf{v} = \mathbf{u}^{[i]} + \sigma D R_{\omega^{[j]}} G_w (2\mathbf{x}^{[i+\frac{1}{2}]} - \mathbf{x}^{[i]})$ 
 $\mathbf{u}^{[i+\frac{1}{2}]} = \mathbf{v} - \text{soft}_\lambda(\mathbf{v})$ 
 $\mathbf{x}^{[i+1]} = \mathbf{x}^{[i]} + \alpha^{[i]}(\mathbf{x}^{[i+\frac{1}{2}]} - \mathbf{x}^{[i]})$ 
 $\mathbf{u}^{[i+1]} = \mathbf{u}^{[i]} + \alpha^{[i]}(\mathbf{u}^{[i+\frac{1}{2}]} - \mathbf{u}^{[i]})$ 
end for
 $\hat{\mathbf{x}}^{[j+1]} = \mathbf{x}^{[I]}$ 
end for

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### IV. EXPERIMENTS AND RESULTS

#### A. Main Experiment

Testing was performed on ten audio signals from EBU SQAM database<sup>1</sup>. The set of signals was chosen to align with previous research in audio inverse problems [1], [4], [5] and are sampled at 44 100 Hz.

Each signal was first peak-normalized and subsequently quantized using a mid-riser uniform quantizer, consistent with [4]. The signals were then truncated to remove initial silence and to ensure equal length across all samples. The resulting duration was 7.1 seconds (i.e., 312 800 samples). Only the first channel of stereo recordings was used in experiments. The experiments were performed on quantized signals with word lengths of 3, 4, 5, 6, 7, and 8 bps.

The following DGT parameters were used in the experiments: window length  $w = 8192$  samples, Hann window, hop size  $a = w/4 = 2048$  samples, and  $M = 8192$  frequency channels. The optimization parameters were set to  $\tau = 0.25$ ,  $\sigma = \alpha = 1$ , and  $\lambda = 0.01$ . The number of iterations was  $I = 50$  for the inner loop, and  $J = 10$  for the outer loop.

For the purpose of quantitative comparison, the sparsity prior analysis (SPA) algorithm from [4] was used. This algorithm is based on formulation (4). The SPA algorithm was tested on the same set of signals, using the parameter settings reported in [4].

The results are shown in Fig. 3 and Fig. 4. The proposed algorithm performs marginally better according to the perceptual metric while the SPA algorithm achieves higher  $\Delta\text{SDR}$  values. Informal listening tests suggest that the proposed method retains more higher frequency content at the cost of audible artefacts, whereas the SPA behaves more like a low-pass filter.

All experiments were performed in MATLAB R2025a. The implementation is provided in the GitHub repository<sup>2</sup> with dependencies on the implementation provided in [1].

<sup>1</sup><https://tech.ebu.ch/publications/sqamed>

<sup>2</sup>[https://github.com/Jan-Fridrich/phase\\_aware\\_dequantization](https://github.com/Jan-Fridrich/phase_aware_dequantization)

## B. Metrics

Signal-to-distortion ratio (SDR) is a measure comparing the original and a distorted (degraded) signal defined as

$$\text{SDR}(\mathbf{x}, \mathbf{y})[\text{dB}] = 10 \log_{10} \frac{\|\mathbf{x}\|_2^2}{\|\mathbf{x} - \mathbf{y}\|_2^2}. \quad (18)$$

This metric represents the relative similarity of the signals. To express the improvement in the dequantized signal,  $\Delta\text{SDR}$  is used as

$$\Delta\text{SDR}_{\mathbf{x}}(\mathbf{y}, \mathbf{z}) = \text{SDR}(\mathbf{x}, \mathbf{y}) - \text{SDR}(\mathbf{x}, \mathbf{z}), \quad (19)$$

where  $\mathbf{x}$  is the original (ground truth) signal and  $\mathbf{y}, \mathbf{z}$  are the dequantized and quantized signals, respectively.

PEMO-Q (v1.4.1), is a method for objective evaluation of perceived audio quality with a scale ODG (objective difference grade) from  $-4$  to  $0$ , where higher is better [11].

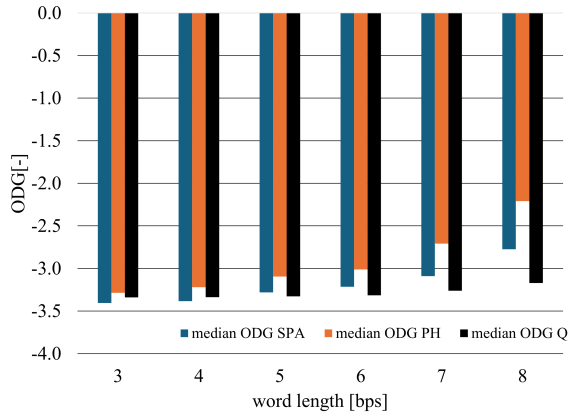


Fig. 3. Median ODG scores of the proposed algorithm (PH) and the SPA algorithm. The Q values correspond to the quantized signal.

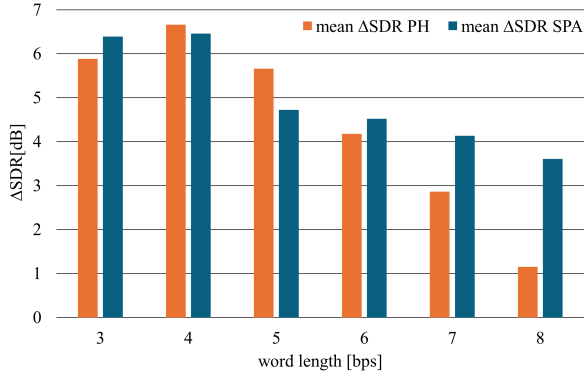


Fig. 4. Mean  $\Delta\text{SDR}$  of the proposed algorithm (PH) and the SPA algorithm.

## C. Parameter $\lambda$

The choice of parameter  $\lambda$  determines the threshold of the soft thresholding operator (see Eq. (17)) and greatly impacts the performance of the algorithm. Tests were performed on the same data as in the main experiment, with a fixed word-length of  $w = 6$ . Best results were obtained by setting the parameter  $\lambda = 0.01$  as seen in Fig. 5.

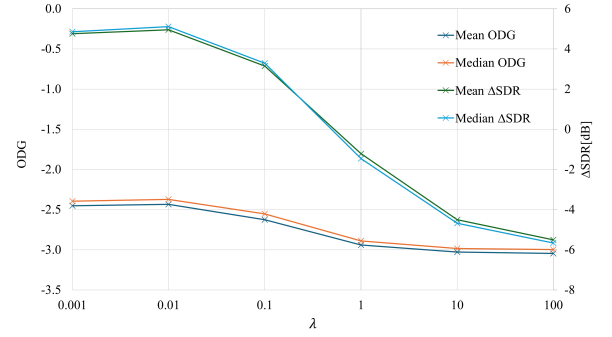


Fig. 5. ODG and  $\Delta\text{SDR}$  versus  $\lambda$ .

## V. CONCLUSION

In this paper, a novel audio dequantization method was proposed. This method, based on [1], exploits the temporal characteristics of a spectrogram with the usual sinusoidality assumption. Quality of the reconstruction was measured by objective metrics  $\Delta\text{SDR}$  and PEMO-Q. Its performance measured against method proposed in [4] is comparable in  $\Delta\text{SDR}$  and slightly better in the perceptual metric PEMO-Q.

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